

A General Rate Theorem for Mixing Measure Estimation

1 The Meta-Theorem

Main Result

Theorem 1.1 (General Rate Theorem). *Let $\theta^* \in \Theta \subseteq \mathcal{M}_+(\mathcal{S}_+^d)$. Suppose the model $\{K_\theta : \theta \in \Theta\}$ has **local metric entropy** around θ^* satisfying:*

$$\log N(\epsilon, \{\theta \in \Theta : d_K(\theta, \theta^*) \leq \delta\}, d_K) \leq D(\theta^*) \cdot \log\left(\frac{\delta}{\epsilon}\right) \quad (1)$$

for all $0 < \epsilon < \delta$ sufficiently small. Then:

$$d_K(\hat{\theta}_m, \theta^*) = O_p\left(m^{-\frac{1}{2+D(\theta^*)/s}}\right) \quad (2)$$

where s is the smoothness parameter if θ^* has a density, or $s = \infty$ (giving rate $m^{-1/2}$) in the parametric case.

More precisely, with $D = D(\theta^*)$:

$$d_K(\hat{\theta}_m, \theta^*) = O_p\left(m^{-\alpha(\theta^*)}\right), \quad \alpha(\theta^*) = \frac{1}{2 + D(\theta^*)} \quad (3)$$

when the density has regularity $s = 1$.

Definition 1.2 (Effective Dimension). The **effective dimension** of θ^* relative to the model Θ is:

$$D(\theta^*) := \lim_{\epsilon \rightarrow 0} \frac{\log N(\epsilon, \Theta_{\theta^*}, d_K)}{\log(1/\epsilon)} \quad (4)$$

where Θ_{θ^*} is a local neighborhood of θ^* in Θ , and $N(\epsilon, \cdot, d_K)$ is the ϵ -covering number in the kernel metric $d_K(\theta_1, \theta_2) = \|K_{\theta_1} - K_{\theta_2}\|$.

Remark 1.3. The effective dimension $D(\theta^*)$ captures the **local complexity** of the model around θ^* . It can vary with θ^* —this is the key to adaptation.

2 The Exponent Formula

Rate Exponent

For θ^* with regularity s (number of derivatives of density, or $s = \infty$ for parametric):

$$\alpha(\theta^*) = \frac{s}{2s + D(\theta^*)} \quad (5)$$

giving rate $d_K(\hat{\theta}_m, \theta^*) = O_p(m^{-\alpha(\theta^*)})$.

Theorem 2.1 (Explicit Exponent Formula). *Let θ^* belong to a class Θ characterized by:*

- **Intrinsic dimension:** $\dim(\text{supp}(\theta^*)) = D_{\text{int}}$
- **Smoothness:** density $p^* = d\theta^*/d\mu$ has s derivatives
- **Sparsity:** θ^* is supported on J atoms (if discrete)

Then:

$$\alpha(\theta^*) = \begin{cases} \frac{1}{2} & \text{if } \theta^* \text{ is } J\text{-sparse (parametric)} \\ \frac{s}{2s + D_{\text{int}}} & \text{if } \theta^* \text{ has } C^s \text{ density on } D_{\text{int}}\text{-dim support} \end{cases} \quad (6)$$

3 Recovering All Special Cases

Proposition 3.1 (Unification). *All previously stated rates are special cases of Theorem ??:*

Structure	$D(\theta^*)$	$\alpha = \frac{s}{2s+D}$	Rate
J -sparse	0 (parametric)	$\frac{1}{2}$	$m^{-1/2} \cdot \sqrt{Jd^2}$
C^s on $\mathcal{S}_+^d$	$\frac{d(d+1)}{2s}$	$\frac{s}{2s+d(d+1)/2}$	$m^{-\frac{s}{2s+d(d+1)/2}}$
C^s on rank- r	$\frac{rd}{s}$	$\frac{s}{2s+rd}$	$m^{-\frac{s}{2s+rd}}$
C^s on rank-1	$\frac{d}{s}$	$\frac{s}{2s+d}$	$m^{-\frac{s}{2s+d}}$
(ϵ, J) -approx. sparse	$\frac{1}{\alpha}$	α	$m^{-\alpha}$

Proof. We verify each case:

Case 1: J -sparse. The parameter space is $\mathbb{R}_+^J \times (\mathcal{S}_{++}^d)^J$ with dimension $p = J(1+d(d+1)/2) \approx Jd^2$. Local covering number: $N(\epsilon, B_\delta, d_K) \lesssim (\delta/\epsilon)^p$, so $D = 0$ in the sense that $\log N \sim p \log(1/\epsilon)$ is constant in the exponent. The rate is $m^{-1/2} \cdot \sqrt{p} = m^{-1/2} \cdot \sqrt{Jd^2}$.

Case 2: C^s density on $\mathcal{S}_+^d$. The intrinsic dimension is $D_{\text{int}} = d(d+1)/2$. By standard metric entropy for C^s functions on D_{int} -dimensional domains: $\log N(\epsilon) \sim \epsilon^{-D_{\text{int}}/s}$. Thus $D(\theta^*) = D_{\text{int}}/s$ and $\alpha = s/(2s + D_{\text{int}})$.

Case 3: Rank- r support. The manifold $\mathcal{S}_{\leq r}^d$ has dimension $rd - \binom{r}{2} + r \approx rd$. So $D_{\text{int}} = rd$ and $\alpha = s/(2s + rd)$.

Case 4: Approximate sparsity with α -decay. If $\inf_{\nu \in \Theta_J} d_K(\theta^*, \nu) \lesssim J^{-\alpha}$, then balancing bias $J^{-\alpha}$ against variance $\sqrt{Jd^2/m}$ gives optimal $J \asymp m^{1/(2\alpha+1)}$ and rate $m^{-\alpha/(2\alpha+1)}$. This corresponds to effective dimension $D = 1/\alpha - 2$. $\square$

4 The General Statement

Most General Form

Theorem 4.1 (Master Theorem). *Let $\theta^* \in \mathcal{M}_+(\mathcal{S}_+^d)$ and define:*

$$D(\theta^*) := \inf \{D \geq 0 : \exists \Theta \ni \theta^* \text{ with } \log N(\epsilon, \Theta, d_K) \lesssim \epsilon^{-D}\} \quad (7)$$

*the **metric entropy dimension** of the smallest model containing θ^* .*

Then the minimax rate over any class $\Theta \ni \theta^$ with entropy dimension D is:*

$$\inf_{\hat{\theta}} \sup_{\theta \in \Theta} \mathbb{E} \left[d_K(\hat{\theta}, \theta)^2 \right] \asymp m^{-\frac{2}{2+D}} \quad (8)$$

If θ^ has a density with s derivatives, replace D with D/s :*

$$\text{Rate} = m^{-\frac{2s}{2s+D_{\text{int}}}} \quad (9)$$

where $D_{\text{int}} = \dim(\text{supp}(\theta^))$.*

Remark 4.2 (Interpretation). The exponent $\alpha(\theta^*) = s/(2s + D_{\text{int}})$ interpolates between:

- $\alpha = 1/2$ when $D_{\text{int}} = 0$ (parametric/sparse)
- $\alpha \rightarrow 0$ as $D_{\text{int}} \rightarrow \infty$ (curse of dimensionality)
- $\alpha \rightarrow 1/2$ as $s \rightarrow \infty$ (infinite smoothness recovers parametric rate)

5 Adaptive Estimation

Theorem 5.1 (Adaptive Rate). *Consider the penalized estimator:*

$$\hat{\theta}_m = \arg \min_{\theta \in \bigcup_{k=1}^{\infty} \Theta_k} \{-\ell_m(\theta) + \text{pen}(\Theta_k)\} \quad (10)$$

where $\{\Theta_k\}$ is a nested sequence of models with increasing complexity (e.g., Θ_k = measures supported on $\leq k$ atoms).

If $\text{pen}(\Theta_k) \asymp D_k \log m$ where D_k = effective dimension of Θ_k , then:

$$d_K(\hat{\theta}_m, \theta^*) = O_p \left(m^{-\alpha(\theta^*)} (\log m)^\beta \right) \quad (11)$$

where $\alpha(\theta^)$ is the **oracle rate** for the smallest model containing θ^* , achieved **without knowing** $D(\theta^*)$ or s .*

Remark 5.2. This is **adaptation**: the estimator automatically achieves the rate corresponding to the true complexity of θ^* , paying only a logarithmic price.

6 Equivalent Characterizations of $D(\theta^*)$

The effective dimension $D(\theta^*)$ admits several equivalent characterizations:

Theorem 6.1 (Equivalent Definitions). *The following are equivalent (up to constants):*

(i) **Metric entropy:**

$$D = \lim_{\epsilon \rightarrow 0} \frac{\log N(\epsilon, \Theta, d_K)}{\log(1/\epsilon)} \quad (12)$$

(ii) **Gaussian complexity:**

$$D = \lim_{\epsilon \rightarrow 0} \frac{\log \mathbb{E} \left[\sup_{\theta: d_K(\theta, \theta^*) \leq \epsilon} G_\theta \right]^2}{\log(1/\epsilon)} \quad (13)$$

where G_θ is a Gaussian process indexed by Θ .

(iii) **Local Rademacher complexity:**

$$D = \lim_{\epsilon \rightarrow 0} \frac{\log \mathcal{R}_m(\{\theta : d_K(\theta, \theta^*) \leq \epsilon\})}{\log(1/\epsilon)} \quad (14)$$

(iv) **Minkowski dimension:**

$$D = \dim_{\text{Mink}}(\text{supp}(\theta^*)) / s \quad (15)$$

when θ^* has a C^s density on its support.

(v) **Approximation-theoretic:**

$$D = \inf \left\{ D' : \inf_{\nu \in \Theta_J} d_K(\theta^*, \nu) \lesssim J^{-1/D'} \text{ for finite mixtures } \Theta_J \right\} \quad (16)$$

7 Information-Theoretic Formulation

Theorem 7.1 (Information-Theoretic Rate). *Let $I(\theta^*; \epsilon)$ denote the mutual information between θ and $(f_1, \dots, f_m)$ when θ is uniform on the ϵ -ball around θ^* . The minimax rate ϵ_m^* satisfies:*

$$m \cdot I(\theta^*; \epsilon_m^*) \asymp \log N(\epsilon_m^*, \Theta, d_K) \quad (17)$$

Solving for ϵ_m^* gives:

$$\epsilon_m^* \asymp m^{-\frac{1}{2+D(\theta^*)}} \quad (18)$$

Remark 7.2. This is the **Yang-Barron** (1999) information-theoretic characterization: the rate is determined by balancing the information gained per sample against the entropy of the parameter space.

8 Summary: The Rate Formula

The Universal Rate Formula

For $\theta^* \in \mathcal{M}_+(\mathcal{S}_+^d)$, define:

$$D_{\text{int}}(\theta^*) = \text{intrinsic dimension of } \text{supp}(\theta^*) \quad (19)$$

$$s(\theta^*) = \text{smoothness (derivatives) of density } d\theta^*/d\mu \quad (20)$$

$$J(\theta^*) = \text{number of atoms if discrete} \quad (21)$$

The rate exponent is:

$$\alpha(\theta^*) = \begin{cases} \frac{1}{2} & \theta^* \text{ is } J\text{-sparse} \\ \frac{s}{2s + D_{\text{int}}} & \theta^* \text{ has } C^s \text{ density} \end{cases} \quad (22)$$

The rate is:

$$d_K(\hat{\theta}_m, \theta^*) = O_p\left(m^{-\alpha(\theta^*)}\right) \quad (\text{up to } \sqrt{\cdot} \text{ and log factors}) \quad (23)$$

The prefactor depends on:

- Jd^2 for sparse (number of parameters)
- $\|p^*\|_{C^s}$ for smooth (regularity constant)
- Separation Δ for mixtures (identifiability)

9 References

- [1] Y. Yang and A. Barron. Information-theoretic determination of minimax rates of convergence. *Annals of Statistics*, 27(5):1564–1599, 1999.
- [2] S. Ghosal, J.K. Ghosh, and A.W. van der Vaart. Convergence rates of posterior distributions. *Annals of Statistics*, 28(2):500–531, 2000.
- [3] L. Birgé. Model selection via testing: an alternative to (penalized) maximum likelihood estimators. *Annales de l'IHP Probabilités et Statistiques*, 42(3):273–325, 2006.
- [4] A.B. Tsybakov. *Introduction to Nonparametric Estimation*. Springer, 2009.
- [5] S. Ghosal and A.W. van der Vaart. *Fundamentals of Nonparametric Bayesian Inference*. Cambridge University Press, 2017.
- [6] A. Rakhlin, K. Sridharan, and A.B. Tsybakov. Empirical entropy, minimax regret and minimax risk. *Bernoulli*, 23(2):789–824, 2017.